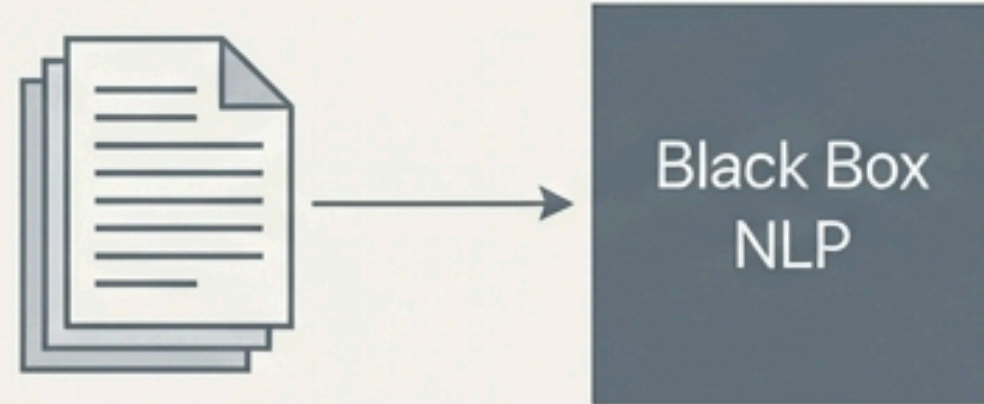


THE JOKE

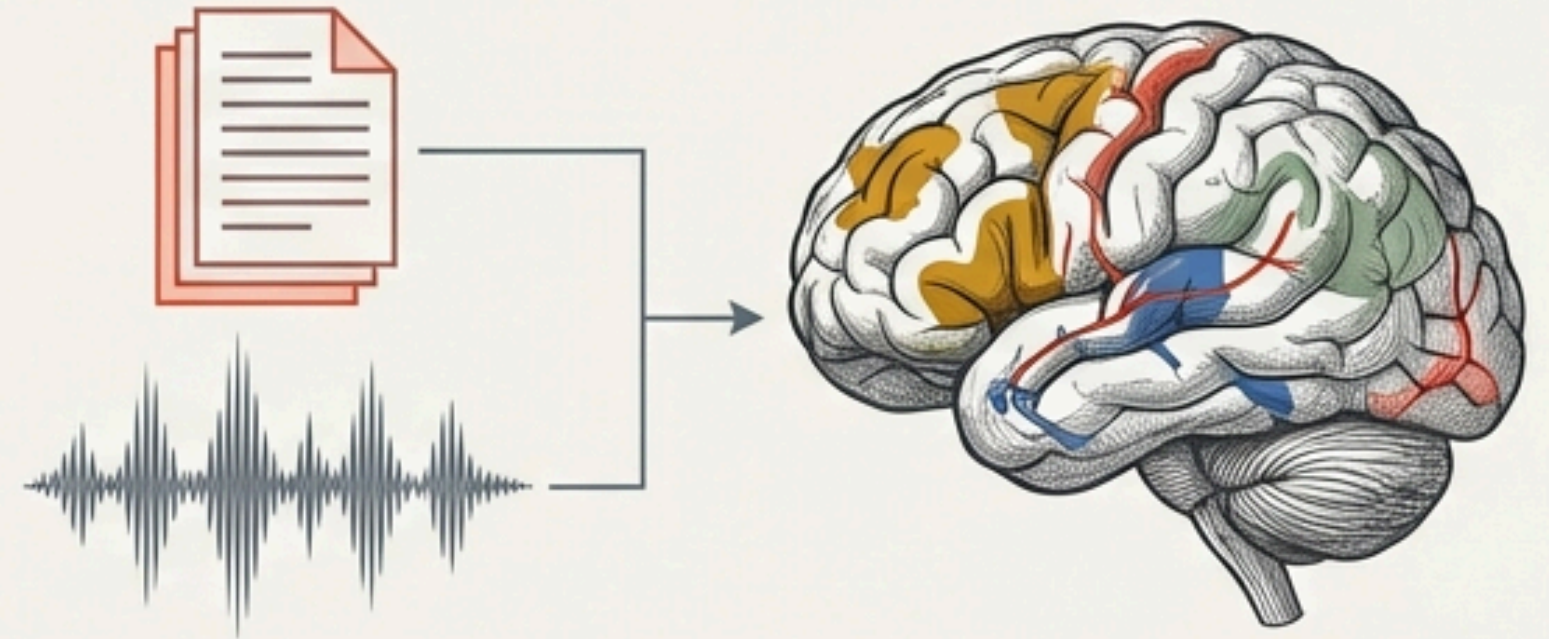
This project predicts the funniness of jokes by bridging NLP and computational neuroscience. Text jokes are converted to speech via TTS and passed through TRIBE v2 (Transformer-based fMRI Brain Encoder), a pre-trained model that simulates neural responses across 20,484 brain vertices. Activations are extracted from four humor-relevant regions of interest — the Temporal-Parietal Junction (TPJ), Middle Temporal Gyrus (MTG), Orbitofrontal Cortex (OFC), and Broca's Area (BA45) — and combined into a weighted funniness score using a regression model

MT24AAI200 - Rohan DVSS
MT24AAI174 - Anjaneya Prasad

Redefining Humor from Linguistic Opinion to Neural Cognition

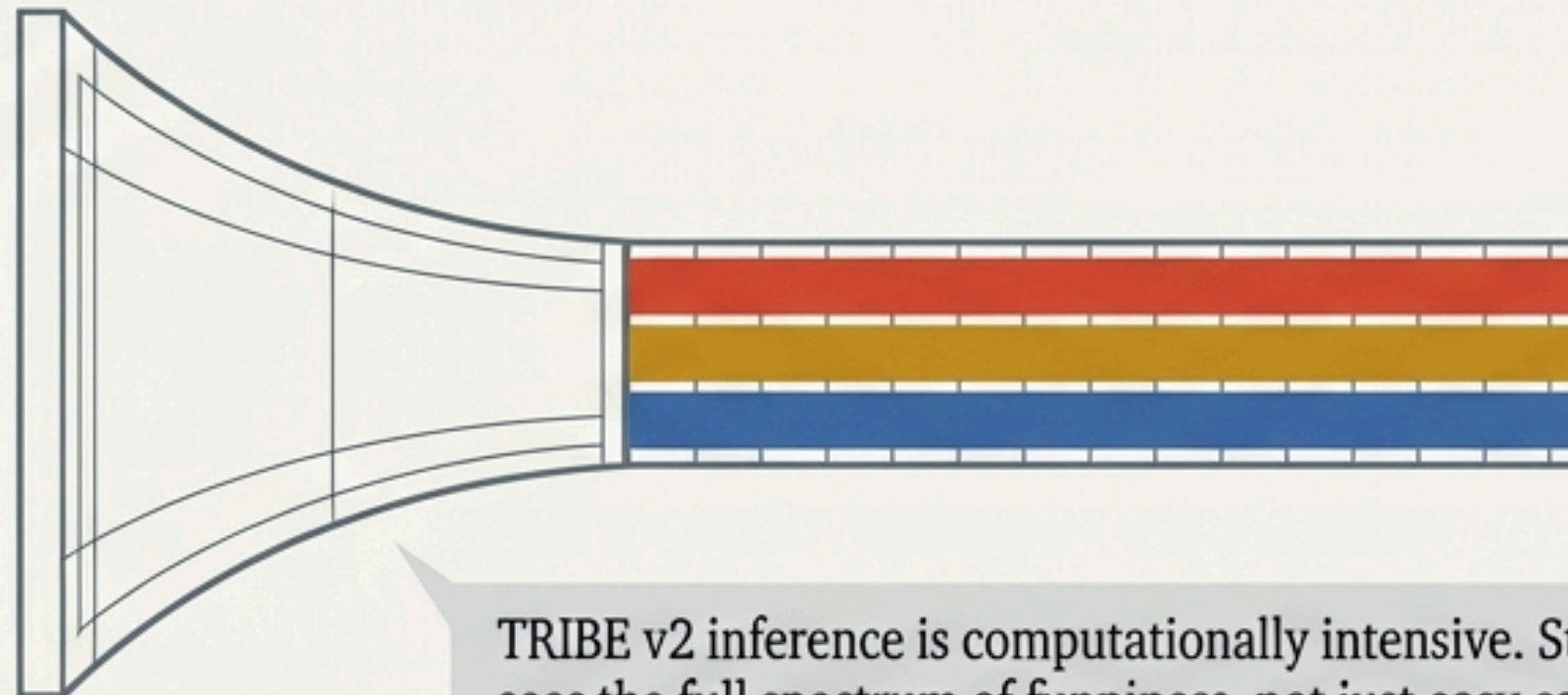


Traditional NLP models what people say is funny. They capture linguistic patterns, measuring opinion.



JokeScore predicts actual brain activation. By mapping neural activity directly to subjective perception, we measure cognition.

200k Total Reddit Jokes



250-300 Stratified Sample
run through TRIBE v2

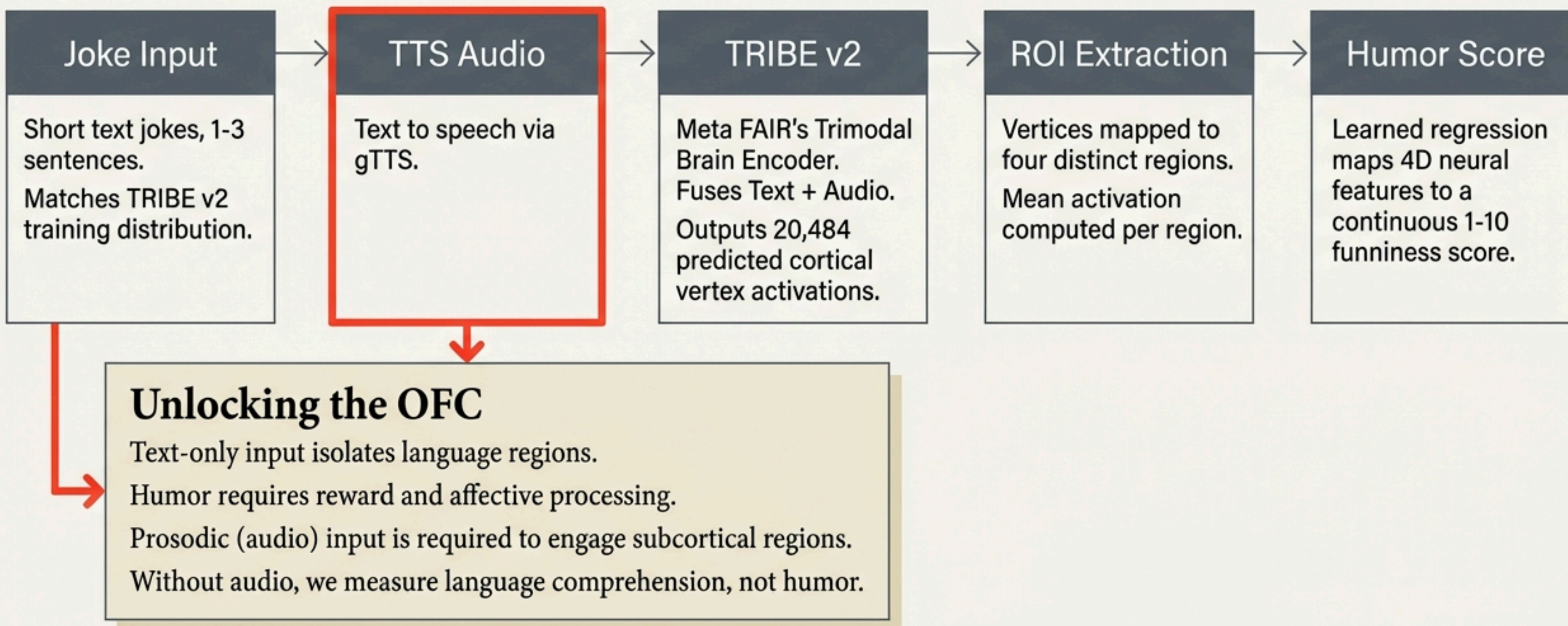
~100 High-score jokes

~75 Mid-range jokes

~100 Low-score jokes

TRIBE v2 inference is computationally intensive. Stratified sampling ensures the regression model sees the full spectrum of funniness, not just easy cases, producing reliable coefficient estimates.

The Simulation Engine Translating Multimodal Stimuli into Neural States



Scientifically Grounded Regions of Interest Form the Predictive Architecture

TPJ (Temporo-Parietal Junction)

Weight: 0.40

Theory of Mind. The social inference engine. Fires when subjects model joke-teller intent and get the violated expectation.

MTG (Middle Temporal Gyrus)

Weight: 0.35

Semantic Incongruity. Detects broken patterns. Fires at the exact moment of the click or punchline.

OFC (Orbitofrontal Cortex)

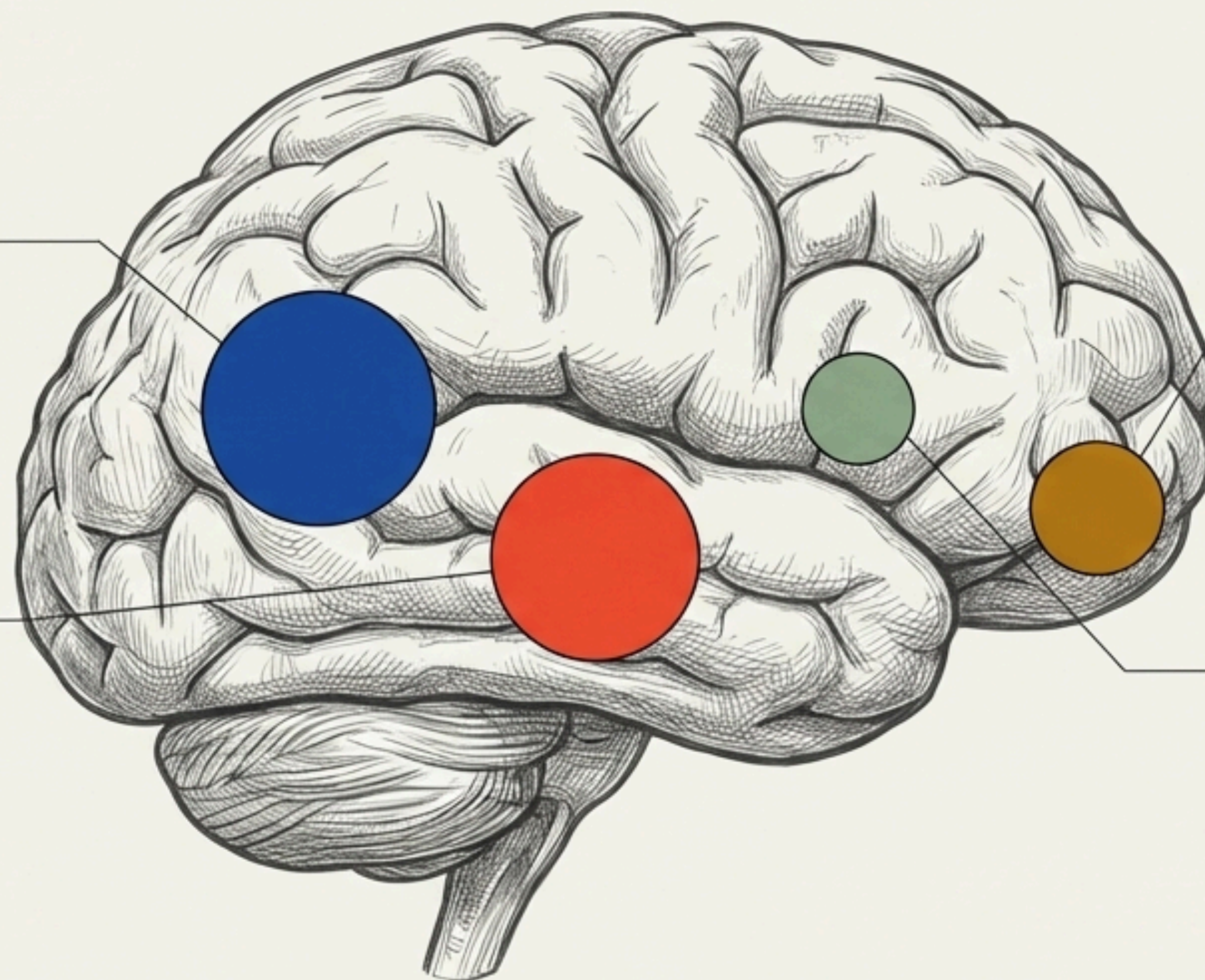
Weight: 0.15

Reward & Pleasure. The affective that's funny signal. Requires audio input to engage.

BA45 (Broca's Area)

Weight: 0.10

Language Complexity. Handles wordplay and syntactic ambiguity.



$$\text{Score} = \beta_0 + (\beta_1 \cdot \text{TPJ}) + (\beta_2 \cdot \text{MTG}) + (\beta_3 \cdot \text{OFC}) + (\beta_4 \cdot \text{BA45})$$

Initial weights serve as a neuroscience-motivated prior. Final weights β_1 - β_4 are learned entirely from data via linear regression. *Relationship is learned, not predefined*

The rJokes Dataset: a Large Scale Humor Collection

Orion Weller and Kevin Seppi

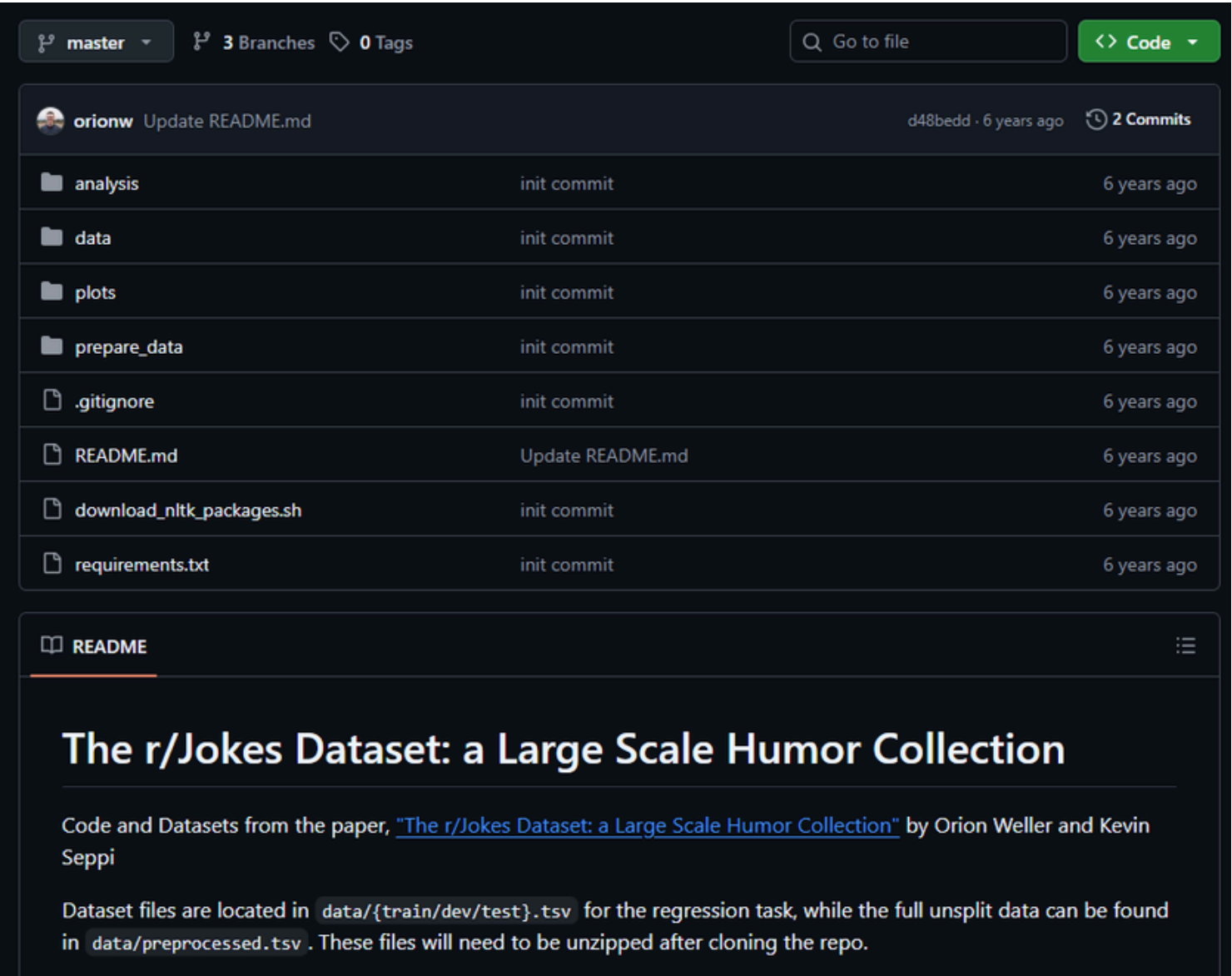
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Abstract

Humor is a complicated language phenomenon that depends upon many factors, including topic, date, and recipient. Because of this variation, it can be hard to determine what exactly makes a joke humorous, leading to difficulties in joke identification and related tasks. Furthermore, current humor datasets are lacking in both joke variety and size, with almost all current datasets having less than 100k jokes. In order to alleviate this issue we compile a collection of over 550,000 jokes posted over an 11 year period on the Reddit r/Jokes subreddit (an online forum), providing a large scale humor dataset that can easily be used for a myriad of tasks. This dataset also provides quantitative metrics for the level of humor in each joke, as determined by subreddit user feedback. We explore this dataset through the years, examining basic statistics, most mentioned entities, and sentiment proportions. We also introduce this dataset as a task for future work, where models learn to predict the level of humor in a joke. On that task we provide strong state-of-the-art baseline models and show room for future improvement. We hope that this dataset will not only help those researching computational humor, but also help social scientists who seek to understand popular culture through humor.

Keywords: Computational Humor, Social Media, rJokes Corpus

#	5	How do you end two deaf persons' arguing? Switch off the light.
1	1	What does a blonde have in common with a hurricane? First there's a lot of suckin and blowin, then you lose your house.
2	2	I went to a white pride parade the other day. The floats just kept going around in circles about 200 miles per hour.
3	1	Georgia and Alabama hate each other Really we have very similar jokes for each other. In Georgia we say that we know the tooth brush was invented in Alabama because if it was invented anywhere else it'd be called a teeth brush. In Alabama they say that we sure have some strange uses for those grout cleaners.
4	0	Going to make an app. I've decided to create an app where you tour museums and art galleries finding famous paintings. I'm going to call it Van Gogh. Source - Me
5	1	The PM of Canada issued the building of a dam The dam was finished and started working, giving the people much needed electricity. Years passed, and the PM eventually got a pet deer which he named Frenklie. When a privatisation wave had recently hit his country, the deer asked him why he wasn't giving the dam for privatisation as well. The deer argued that a private company would be better suited to managing it, and it would benefit the citizens. Was he perhaps somehow attached to this dam? After listening closely, the PM responded, "Frenklie, my deer, I don't give a dam." My gift to you who seem to like these kind of jokes so much
6	3	I'd make a joke about my neighbor.... But it would be too close to home
7	0	What is the difference between God and you Ex-wife? God only wants 10 things. Your ex-wife wants half of everything.
8	0	"I swear, I'll castrate you!!!" "Do it, you won't. No balls bro."
9	4	The first thing a man notices about a woman is her eyes.. And when he made sure it's not looking his way,he checks out her tits ..
10	1	What's your favourite insult joke? I'll get it started: 2090 called, you're dead and nobody misses you
11	2	Trump: "Hows that Mexican mall going?" "Mall? We thought you said wall" Trump: "No way that's harsh, also hows that Muslim band looking?"
12	0	Keanu Reeves donates so much blood... ... that anyone who has ever received donor blood is at least 6% Keanu Reeves.
13	3	What does a nut say when he sneezes? Ca-shew!
14	2	A man decides to ask out the quiet girl from town... He shows up to her home and asks her mother to call for her. The only thing to do in their small town is a carnival in town so after a long quiet drive they make it to the fair. After walking around not talking the man, frustrated, asks her, "What did you have in mind for tonight?" She responds, "well I wanted to get weighed." Just so happens there's a booth there where a man guesses your weight. The man, thrilled just to have the young girl talking, rushes over to the stand. The man at the attraction guesses the girls weight and misses. She collects her prize and the couple walks away. After a long silence the man asked, "well is there anything else you'd like to do?" She responds "I really want to get weighed." Confused, the man walks the girl back to the guessing booth but the man there has had time to size the girl up and guesses her weight on the nose. She receives no prize and fed up with the date storms off. When she returns home her mother stops her and asks, "You're home early, how was your date?" The girl says "wousy"



Thank You.